



Problems (Complex Numbers)

1. For any two complex numbers z_1 and z_2 , prove that $Re(z_1 z_2) = Re(z_1)Re(z_2) - Im(z_1)Im(z_2)$
2. If $z_1 = 2 - i$, $z_2 = 1 + i$, find $\left| \frac{z_1 + z_2 + 1}{z_1 - z_2 + 1} \right|$
3. Let $z_1 = 2 - i$, $z_2 = -2 + i$, find
 - a. $Re\left(\frac{z_1 z_2}{\bar{z}_1}\right)$,
 - b. $Im\left(\frac{1}{z_1 \bar{z}_1}\right)$
4. Find modulus of $\frac{1+i}{1-i} - \frac{1-i}{1+i}$
5. If α, β are different complex numbers with $|\beta| = 1$, find $\left| \frac{\beta - \alpha}{1 - \bar{\alpha}\beta} \right|$
6. Find the non-zero integer solution to the equation $|1 - i|^x = 2^x$
7. Two of the roots of a 4th degree polynomial with real coefficients are given as $(1+i)$ and $(2-i)$. What is the sum of all the roots?
8. Solve $x^3 = 1$. Show that
 - a. the one of the complex roots is square of the other complex root.
 - b. The sum of the roots = 0